



Technical Service Bulletin

Technical Service Bulletin: TSB160018

Released Date: 18-May-2016

Lubricating Oil Cooler O-Ring Replacement

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Warranty Statement

The information in this document authorizes specific changes to the repair practice for failures covered under product warranty coverages.

Contents

Product Affected

- ISX15 (All Versions)
- QSX15 (All Versions)

Issue

Symptom:

- Lubricating oil leak or coolant leak from the threaded plugs and fitting on the lubricating oil cooler.

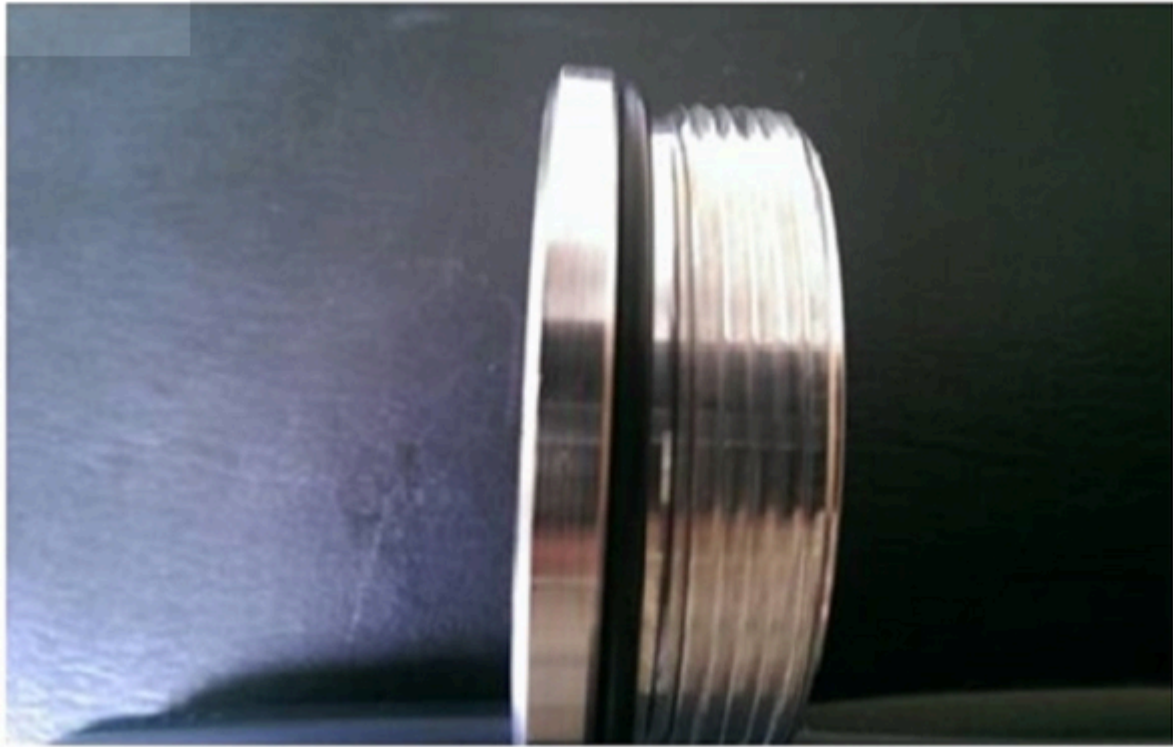
Root Cause:

- Material malfunction of the o-rings.

Verification

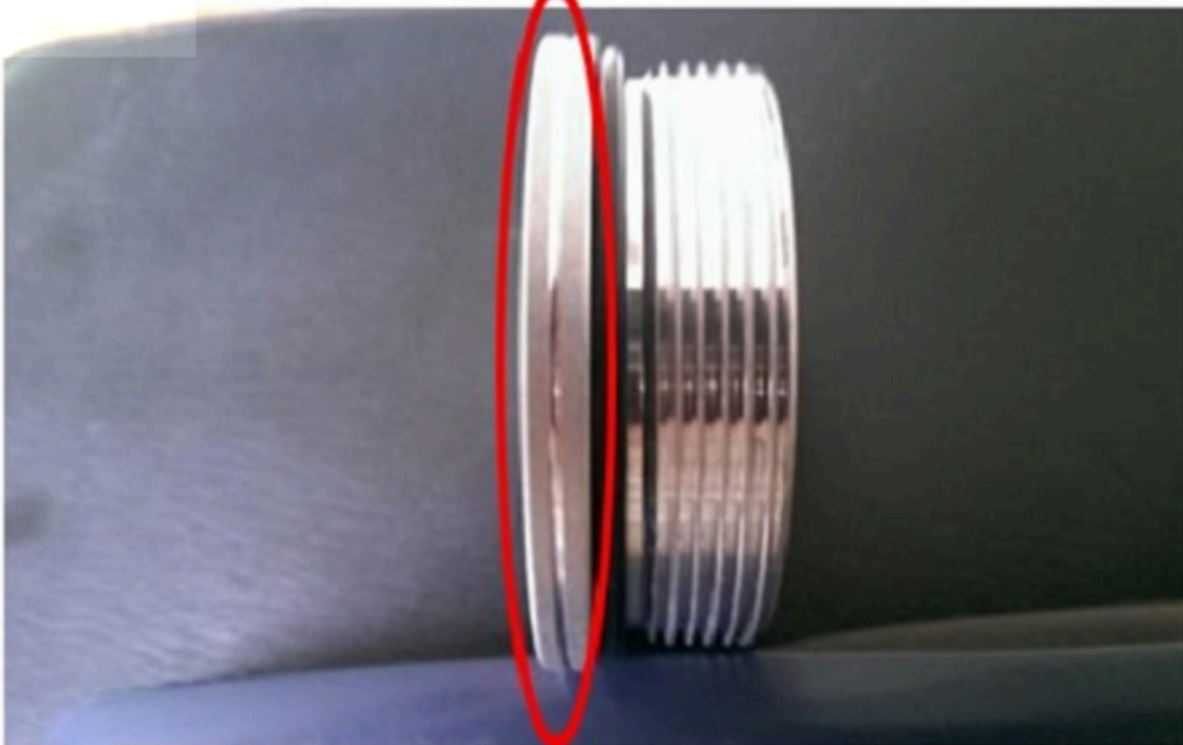
- Verify coolant or lubricating oil is leaking from at least one of the six threaded plugs or fittings shown in Figure 3 below.
- A unit equipped with suspect o-rings can be identified by a groove on the threaded plug for the lubricating oil thermostat. See Figures 1 and 2 below. With the threaded plug still installed, inspect for a groove.
- Proceed to the Resolution section below **only** if a groove is present on the threaded plug and the engine was built prior to March 25, 2015.

Note : Engines built after March 25, 2015 may have a grooved plug but these units have been assembled with the correct O-rings. Verify the engine build date using the physical engine dataplate or the dataplate found on QuickServe® Online.



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Figure 1, Example of Oil Cooler Thermostat Plug with No Groove and Good O-ring.



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Figure 2, Example of Oil Cooler Thermostat Plug with Groove and Suspect O-ring.

Resolution

- If any of the seven threaded plugs or fittings shown in Figure 3 below are leaking

And

- If the threaded plug for the lubricating oil thermostat (Figure 3, Item 1) has a groove present

Resolution:

- All o-rings listed below should be replaced if installed on the lubricating oil cooler housing.
- See Service Parts Availability section and Service Instructions section below.

Service Parts Availability

- Service tool, Part Number 5394435, is being made available free of charge to all dealers and Cummins® distributors to assist in the replacement of the suspect o-ring seal for the lubricating oil thermostat threaded plug. These tools have been shipped to your local distributor branch headquarters and will be distributed to all certified repair locations in that territory. All tools should be distributed by mid-April 2016. Contact your local Cummins® distributor parts manager with questions.
- Seal kit, Part Number 4376561, contains seven revised o-rings, and is available for purchase. If the lubricating oil cooler housing has plug number 7 (Figure 3, Item 7) an additional o ring seal, Part Number 3678912, will be needed and **must** be ordered separately.

Table 1, Seal Kit and Service Tool Part Number

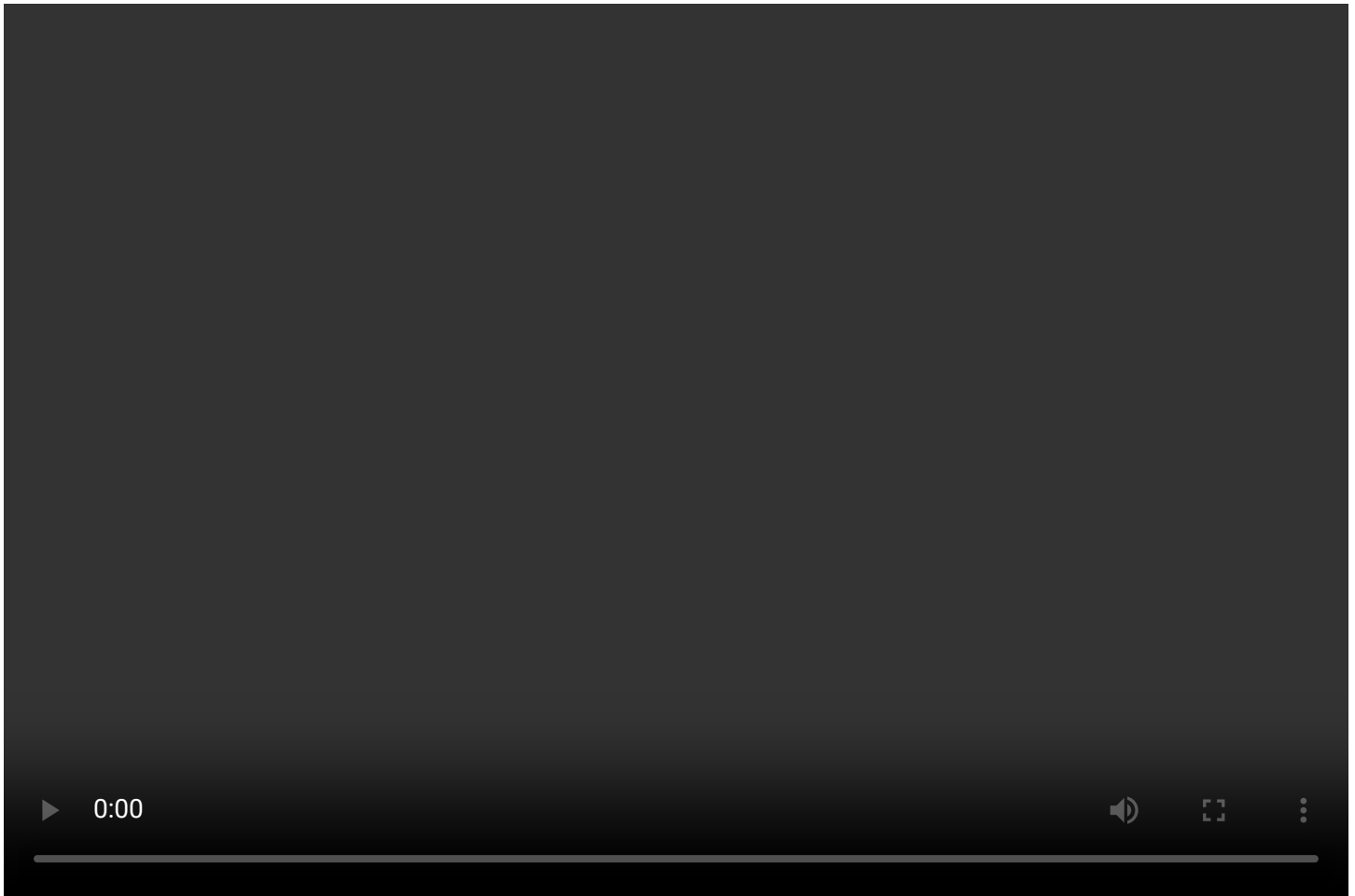
Part Description	Part Number
WRENCH,HEXAGON	5394435
KIT,SEAL	4376561

Table 2, Parts Needed if Seal Kit, Part Number 4376561, is not Available

Part Description	Part Number	Quantity	Location in Figure 3
SEAL,O RING	3089025	1	1
SEAL,O RING	3678606	1	2
SEAL,O RING	3678912	3	3, 4, and 5
SEAL,O RING	3679139	1	6
SEAL,O RING	3922794	1	6

Service Tools and Training Available

Instructional video available. See video below.



Note : If you are not able to see the video window, reference the link below and navigate to the heading, "Computer/Browser/Video Settings That Affect QuickServe® Online" and select "What do I need to check when a document I am viewing indicates a video is available but I am unable to see the video?".

Note : <https://qsol.cummins.com/qs3/qsol/faqs/summary.html>
(<https://qsol.cummins.com/qs3/qsol/faqs/summary.html>)

Personal Protection Equipment (PPE)

- Completing a Job Safety Assessment (JSA) prior to performing work helps identify job safety hazards and prevent incidents. Use the guidelines below to assess if a situation is safe or at risk prior to performing designated work.
- To reduce the possibility of personal injury, personal protective equipment (PPE) must be utilized. Reference General Safety Instructions Procedure 204-006 in Section I of the corresponding Service Manual.

Service Instructions

Select Service Tools

Recommended Cummins® Service Tools

- Hexagon Wrench, Part Number 5394435.

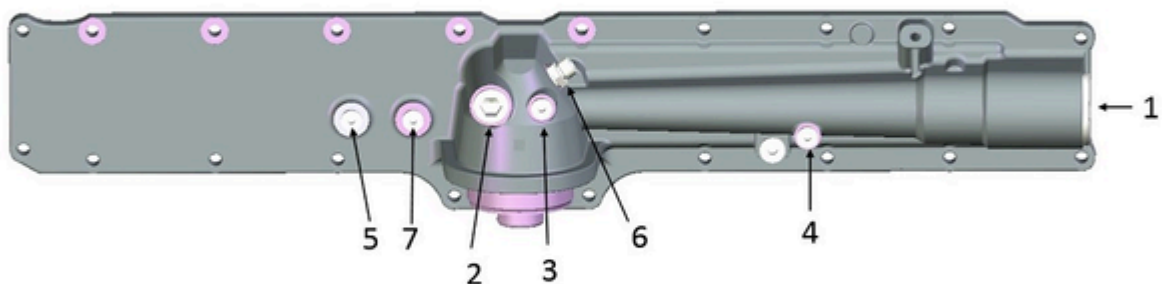
Additional Service Items

- ½ inch ratchet

- White paint marker
- Pry bar or large screwdriver
- Safety glasses
- Gloves
- Scraper
- Brass hammer
- Punch

Component Location

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Figure 3, Plug and Fitting Locations on Lubricating Oil Cooler Housing.

Reference numbers for plugs and fittings to be removed and o-ring locations for this procedure. (See Figure 3 above for component location):

1. Lubricating Oil Thermostat Threaded Plug - Oil
2. Lubricating Oil Filter Bypass Valve Threaded Plug- Oil
3. Threaded Plug Near Lubricating Oil Filter Bypass Valve- Oil
4. Thread Plug Near Lubricating Oil Thermostat- Oil
5. Threaded Plug - Coolant
6. Turbocharger Oil Supply Line Fitting- Oil
7. Optional Threaded Plug - Coolant (This plug will **only** be installed on some units and **only** needs to be replaced if it exists on the housing and a different fitting was **not** installed into this location)

Label each plug or fitting as they are removed. This will help identify the plugs or fitting through the replacement procedure.

Preparatory Steps

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Clean area with solvent and dry with shop air.
- If needed, remove the charging system alternator to access lubricating oil thermostat threaded plug. See the corresponding Service Manual. Reference Procedure 013-001 in Section 13.
- Remove the two capscrews from the water pump housing that extend into the charging system alternator/refrigerant compressor mounting bracket.

Replace

1.) O-ring for Lubricating Oil Thermostat Threaded Plug (Figure 3, Item 1):

- Loosen the threaded plug by hitting the plug with a brass punch. Where the installation allows, this should be done in a forward angle rather than a 90 degree angle to the plug. This will help to prevent the punch from slipping. See Figure A.

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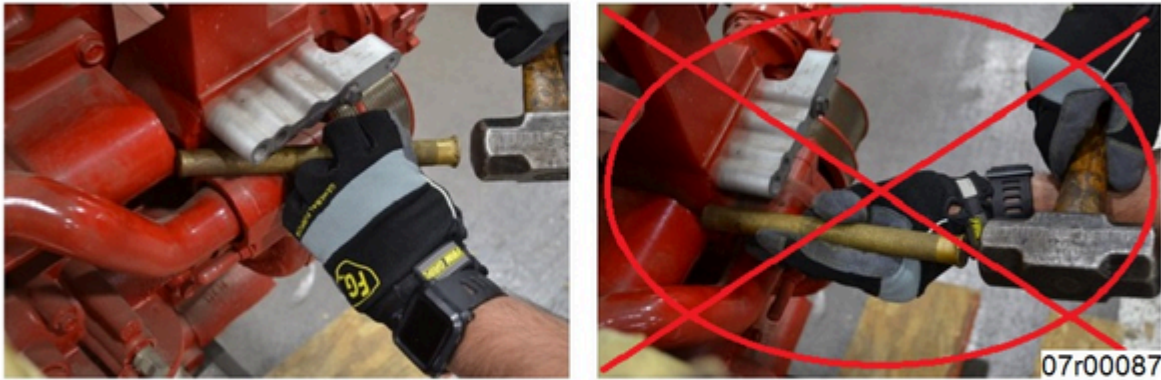


Figure A, Example Showing Correct Forward Angle Verses an Incorrect 90 Degree Angle to the Plug.

- Inspect the threaded plug. If burrs or sharp edges are present, file the area prior to installing the new O-ring.
- Insert tool, Part Number 5394435, into hex feature of lubricating oil thermostat threaded plug and while using a pry bar or large screwdriver to maintain engagement into hex feature of threaded plug. See Figure 4, Item 1 below.

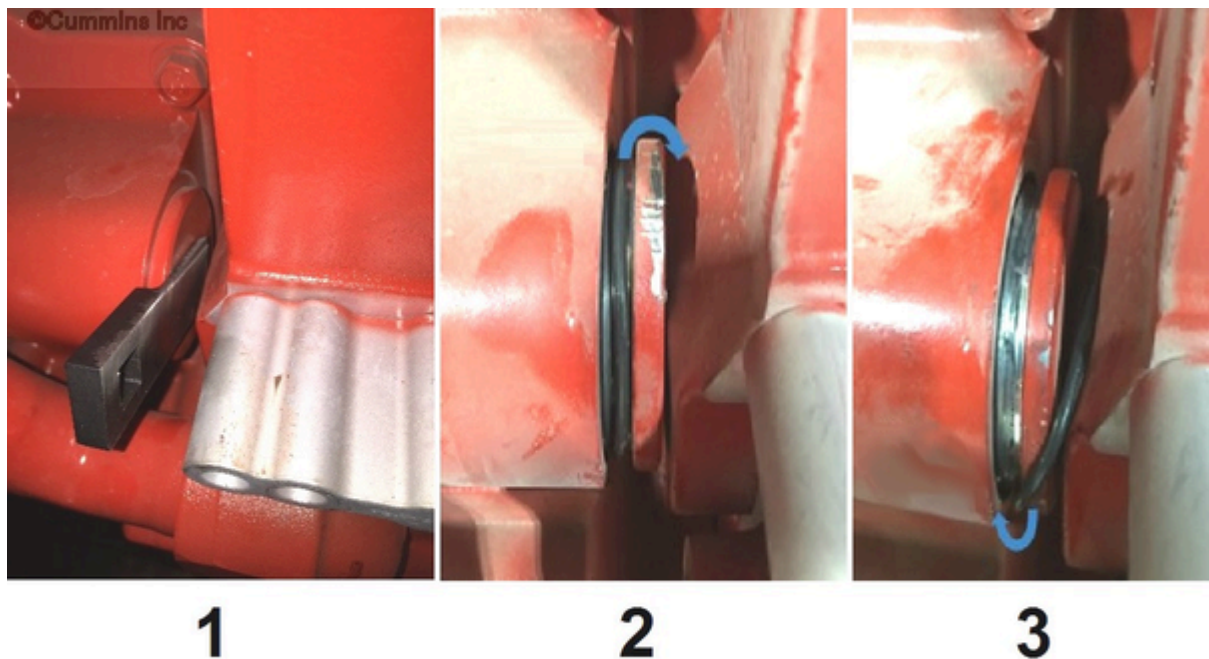
Note : If the tool, Part Number 5394435, does not fit or is not able to loosen the plug, removal of the charging system alternator/refrigerant compressor mounting bracket may be necessary. See the corresponding Service Manual. Reference Procedure 009-056 in Section 9.

- Insert ½ inch ratchet into tool and loosen threaded plug. Remove tool from threaded plug.
- Rotate threaded plug **counterclockwise** by hand until sufficient room exists for removal of suspect o-ring.
- Remove and discard suspect o-ring. See Figure 4, Item 2 below.
- Rotate threaded plug **counterclockwise** by hand until sufficient room exists to install new o-ring.
- Roll new o-ring, Part number 3089025, onto threaded plug until new o-ring slips into o-ring groove. See Figure 4, Item 3 below.

- Verify new o-ring is properly installed and rotate threaded plug **clockwise** back into plug bore by hand.
- Insert tool into hex feature of threaded plug and use a pry bar or large screwdriver to maintain engagement into hex feature.
- Tighten threaded plug until plug makes full contact with lubricating oil cooler housing.
- Use ½ inch torque wrench to tighten threaded plug while using a pry bar or large screwdriver to maintain engagement.

Torque Value: 90 N•m [66 ft-lb]

- Place a small 'x' to the left of the plug with white paint marker.



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Figure 4, Removal and Installation of O-ring:

1. Tool, Part Number 5394435, Inserted Into Hex Feature of Lubricating Oil Thermostat Plug.
2. Removal of Suspect O-ring.
3. Installation of New O-ring.

2.) O-ring for Lubricating Oil Filter Bypass Valve Threaded Plug (Figure 3, Item 2):

- Remove lubricating oil filter bypass valve threaded plug.
- Remove and discard suspect o-ring.
- Install new o-ring, Part Number 3678606, onto threaded plug.
- Install threaded plug.

Torque Value: 45 N•m [33 ft-lb]

- Place a small 'x' to the left of the plug with white paint marker.

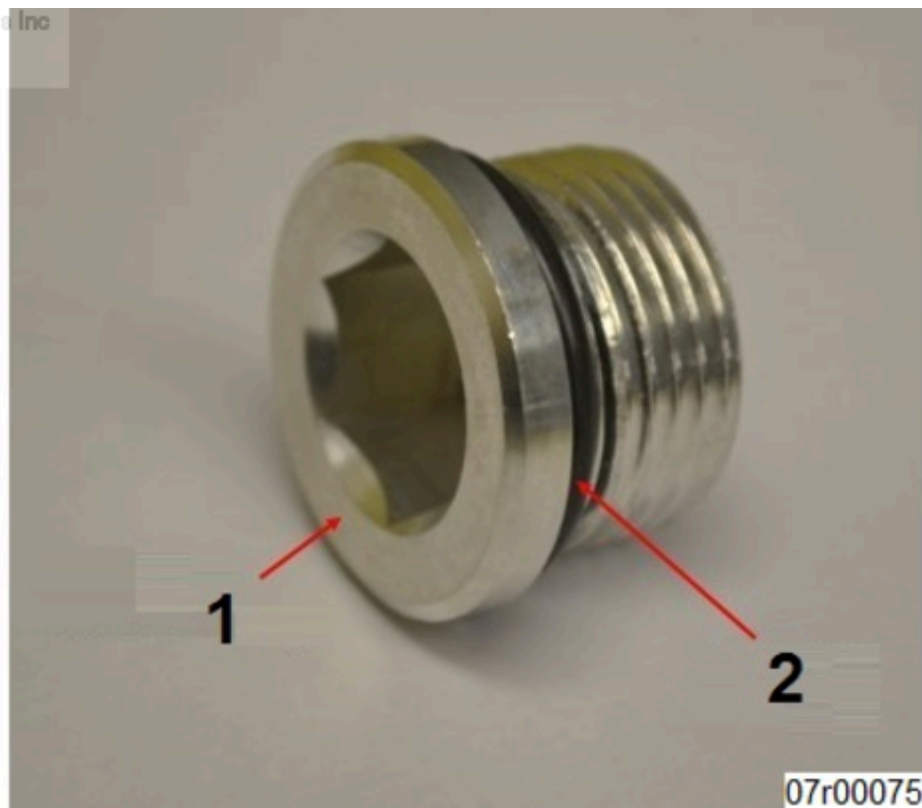


Figure 5, 1.) Lubricating Oil Filter Bypass Valve Threaded Plug. 2.) O-ring.

3.) O-ring for Threaded Plug Near Lubricating Oil Filter Bypass Valve (Figure 3, Item 3):

- Remove threaded plug near lubricating oil filter bypass valve.
- Remove and discard suspect o-ring.
- Install new o-ring, Part Number 3678912, onto threaded plug.
- Install threaded plug.

Torque Value: 12 N•m [9 ft-lb]

- Place a small 'x' to the left of the plug with white paint marker.

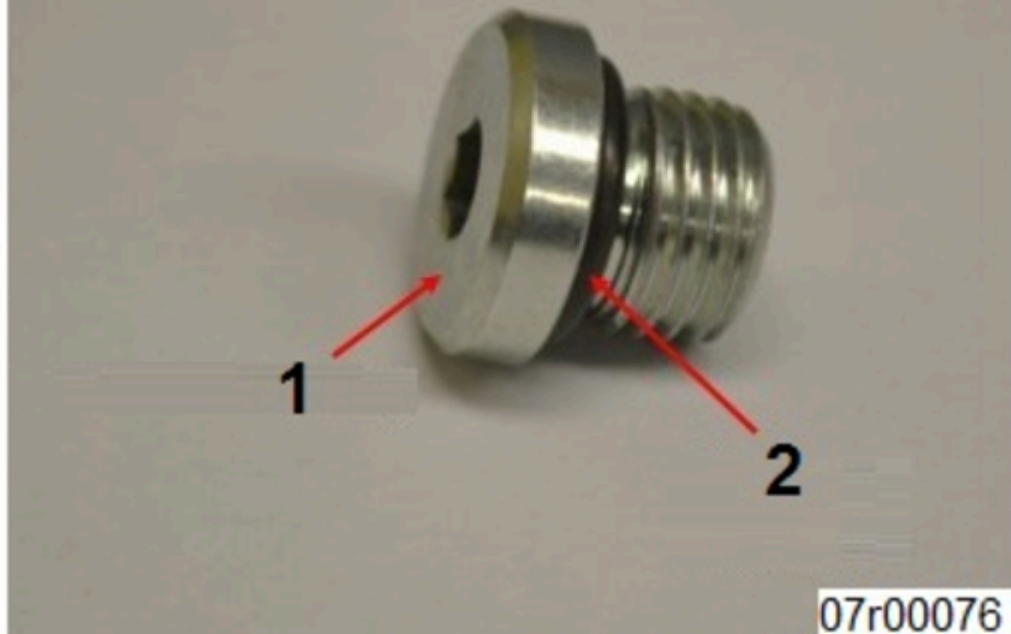


Figure 6, 1.) Threaded Plug Near Lubricating Oil Filter Bypass Valve. 2.) O-ring.

4.) O-ring for Threaded Plug Near Lubricating Oil Thermostat (Figure 3, Item 4):

Note : If a coolant dam or coolant vacuum tool is not available, the threaded plug removed from location number 4 will be used to quickly replace the threaded plug removed from location number 5 to prevent coolant loss.

- Remove threaded plug by lubricating oil thermostat.
- Remove and discard suspect o-ring.
- Install new o-ring, Part Number 3678912, onto threaded plug.

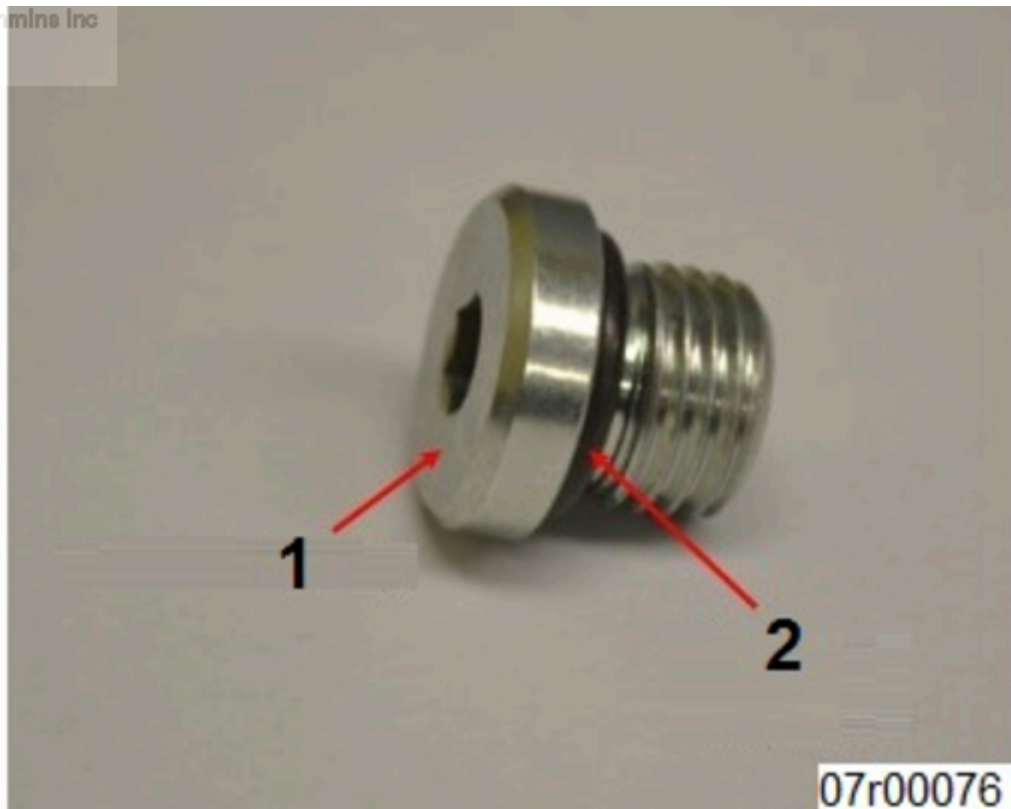


Figure 7, 1.) Threaded Plug Near Lubricating Oil Thermostat. 2.) O-ring.

5.) O-ring for Threaded Plug (Figure 3, Item 5)

- Remove threaded plug number 5 and insert threaded plug number 4 into location threaded plug number 5 was removed from.

Torque Value: 12 N•m [9 ft-lb]

- Remove and discard suspect o-ring from threaded plug number 5.
- Install new o-ring, Part Number 3678912, onto threaded plug.
- Install threaded plug into location threaded plug number 4 was removed from.

Torque Value: 12 N•m [9 ft-lb]

- Place a small 'x' to the left of the plug with white paint marker.

6.) O-rings for Turbocharger Oil Supply Line Fitting (Figure 3, Item 6):

- Remove turbocharger oil supply line from the oil supply line fitting (Figure 9, Item 1).
- Remove the turbocharger oil supply line fitting (Figure 9, Item 2).
- Remove and discard both suspect o-rings.
- Install new o-rings, Part Number 3679139 and 3922794, onto fitting.
- Install the fitting.

Torque Value: 36 N•m [26.5 ft-lb]

- Install turbocharger oil supply line to fitting

Torque Value: 30 N•m [22 in-lb]

- Place a small 'x' to the left of the fitting with white paint marker.

Note : Step 7 below will only be performed on select chassis.

7.) O-ring for Optional Threaded Plug Near Lubricating Oil Filter Bypass Valve (Figure 3, Item 7):

If the Lubricating Oil Cooler Housing has a threaded plug in this location:

- Remove threaded plug near lubricating oil filter bypass valve.
- Remove and discard suspect o-ring.
- Install new o-ring, Part Number 3678912, onto threaded plug.
- Install threaded plug.

Note : If a coolant dam or coolant vacuum tool is not available, the threaded plug removed from location number 3 will be used to quickly replace the threaded plug removed from location number 7 to prevent coolant loss.

Torque Value 12 N•m [9 ft-lb]

- Place a small 'x' to the left of the plug with white paint marker.

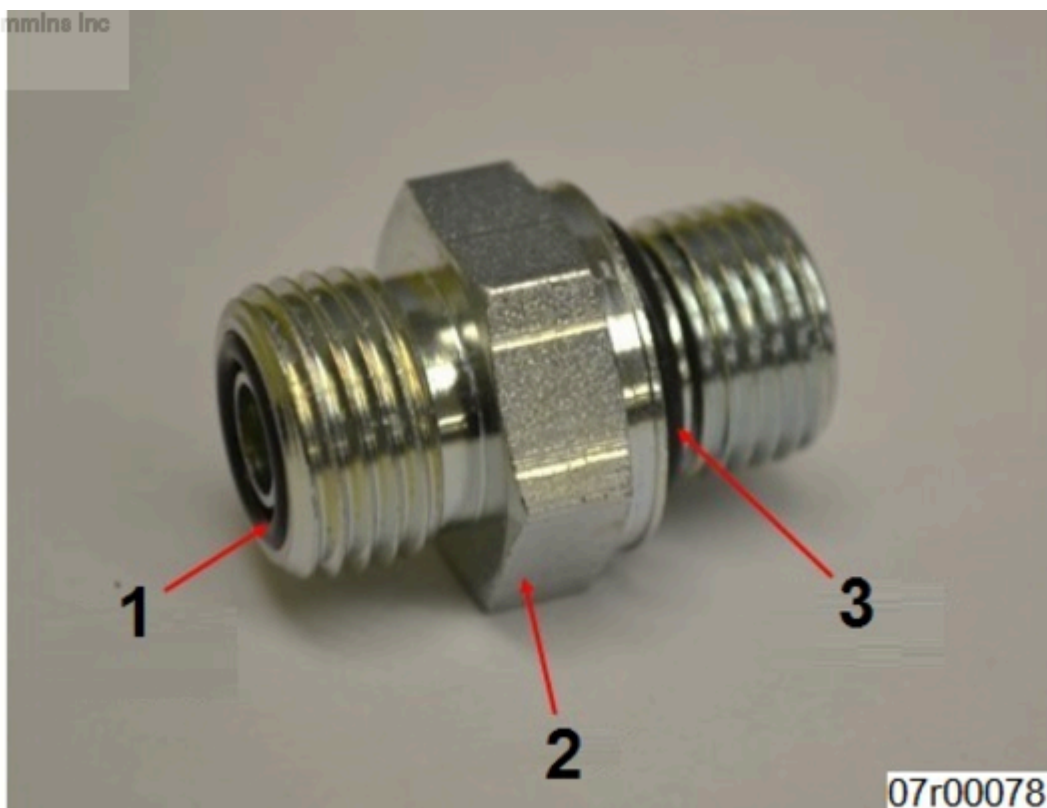


Figure 8, 1.) O-ring. 2.) Turbocharger Oil Supply Line Fitting 3.) O-ring.

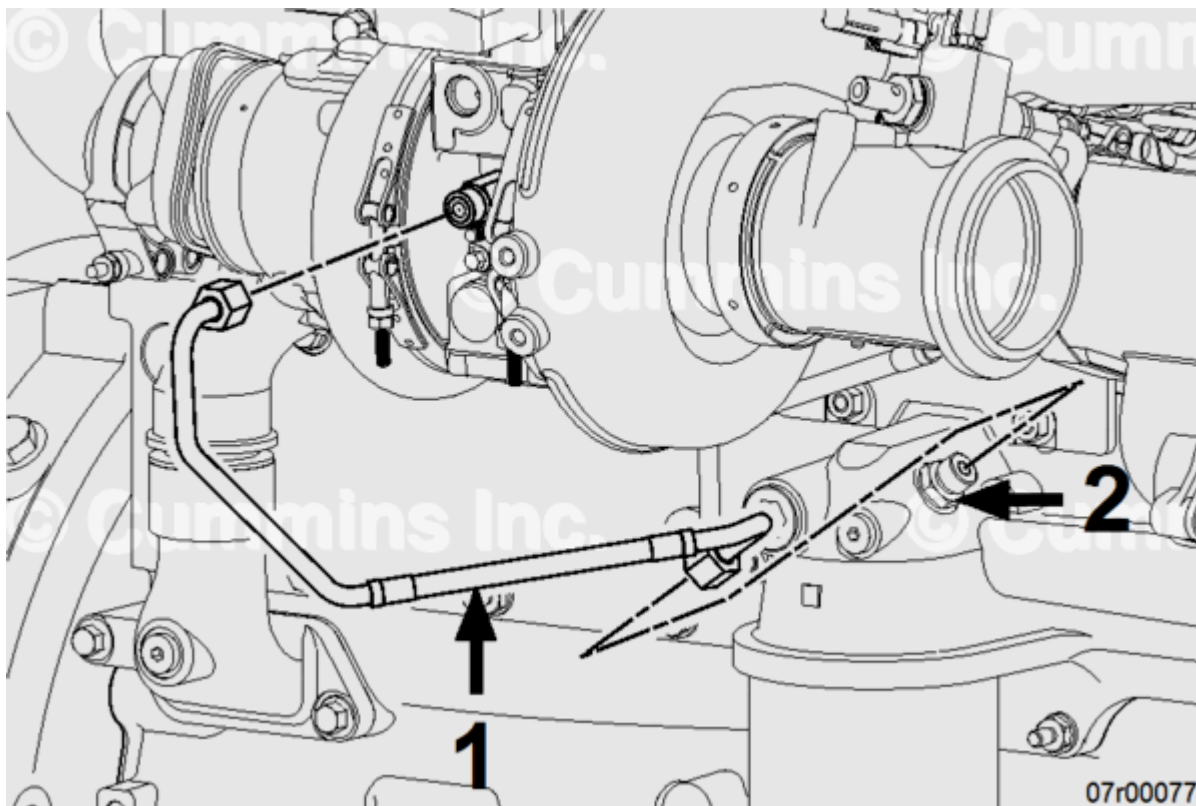


Figure 9, 1.) Turbocharger Oil Supply Line. 2.) Turbocharger Oil Supply Line Fitting.

Finishing Steps

- Install capscrews removed from the water pump housing.

Torque Value: 30 N•m [22 ft-lb]

- Install charging system alternator if removed. See the corresponding Service Manual. Reference Procedure 013-001 in Section 13.
- Install the charging system alternator drive belt. See the corresponding Service Manual. Reference Procedure 013-005 in Section 13.
- Connect the batteries. See equipment manufacturer service information.
- Add coolant to full cold line if necessary. See the corresponding Service Manual. Reference Procedure 008-018 in Section 8.
- Operate the engine to normal operating temperature and check for leaks.

Production Status

Implemented for production. See Table 3.

Engines built after the build date in Table 3 do **not** need to be fitted with new O-rings.

Table 3, Production Information

Product	Engine Serial Number (ESN) First	Build Date*	Engine Plant
QSX15 CM570	79825116	25-Mar-15	Jamestown Engine Plant

*Engine build date can be found on the engine dataplate.

Document History

Date	Details
2016-2-19	Module Created
2016-4-6	Added mention of seventh o-ring, message about service tool, Part Number 5394435, availability, video link, and Step 7.
2016-5-10	Added information about engine built before and after March 25, 2015. Added brass punch and hammer to tools, instructions on how to loosen the plug, and Figure A.
2016-5-17	Updated lubricating oil cooler threaded plug torque value from 99 N•m to 90 N•m.

Last Modified: 18-May-2016
